INDEPENDENT & DEPENDENT VARIABLES

An independent variable is simply a variable whose value does not depend on another. In elementary algebra, it is also typically the input value: the value that changes or determines another value.

If that made no sense: Let's say you're planting a garden, and are trying to maximize the height of a tomato vine. You have fertilizer, water, and other nutrients designed to help your tomato vine grow -- this is the independent variable. How much you add will determine the dependent variable -- in this case, how much your plant grows (the height of the plant is dependent on how much fertilizer you apply).

When graphing functions, we usually denote the independent variable (also called input) as x , and the dependent variable (also called output) as y.

RELATION

A relation is simply a set of inputs and outputs that are related in some way.

Input (x)	Output (y)
1	2
2	3
2	6

Input (x)	Output (y)
1	1
2	4
3	9

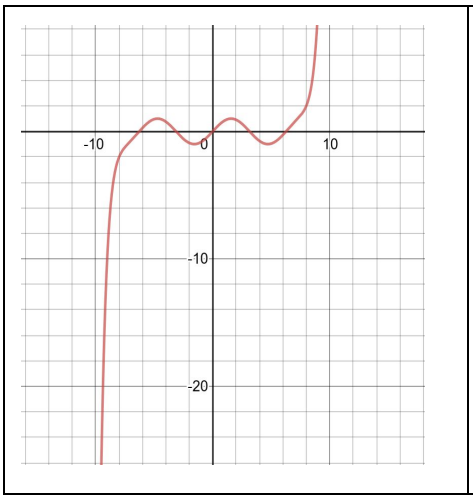
Input (x)	Output (y)
1	5
2	5
3	5

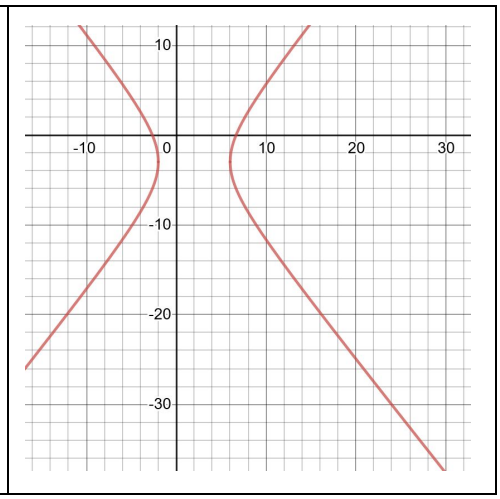
If a relation is such that each input only has 1 unique output, then it's a special type of relation called a *function*.

FUNCTIONS

A function describes a relation between 2 sets, assigning each value in one set (called the domain; i.e. the "input" set) exactly one corresponding value in the other set (called the codomain; i.e. the "output" set). In other words, there is only one y that corresponds to one x.

To determine if a graph is a function, there is a simple method called the Vertical Line Test. You simply draw a straight line parallel to the y-axis. If for the entirety of the graph, there exists only 1 intersection between the line at the graph, then the graph is a function.





ABOVE:

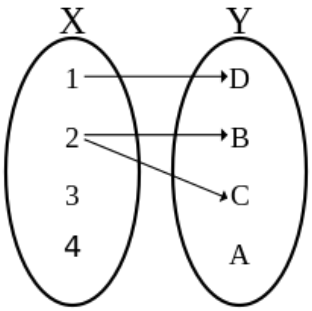
The graph on the light is a function, since for any x on the domain, you can draw a vertical line parallel to the y-axis (x = 0) and it will only intersect the graph at 1 point.

In contrast, the graph on the right is not a function, since there is at least 1 point on the domain such that if you draw a vertical line parallel to the y-axis, the line will intersect the graph at 2 distinct points. Rather, this graph is a *parametric function* (specifically a *hyperbola*), although we will not discuss that here.

On the contrary, it is okay for a function to have more than one element in its domain map to the same codomain. We'll elaborate on this further as you progress.

SIDENOTE: Codomain v. Range

The term "codomain" and "range" are not synonymous -- the codomain is often defined as the range, although some mathematicians define it as the image. For the sake of simplicity, we will not elaborate on this topic, as it requires some knowledge regarding sets, and will instead use the term codomain and range interchangeably.



On the diagram to the right, X represents the DOMAIN and y represents the CODOMAIN. The elements 1,2,3,4 and D,B,C,A

are elements of x and y, respectively. As you can see, 1 “maps” to D, while 2 “maps” to both B and C. Since an element in the domain maps to 2 elements in the codomain, this is not a *function*, but rather a *relation*.

Exercise: Determine if the relation below is a *function*, or just a *relation.* Explain why.

(A)

Input (x)	Output (y)
1	2
2	3
2	6
5	7

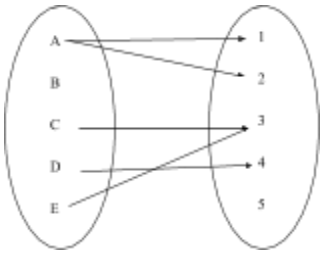
(B)

Input (x)	Output (y)
1	1
2	4
3	9
4	16

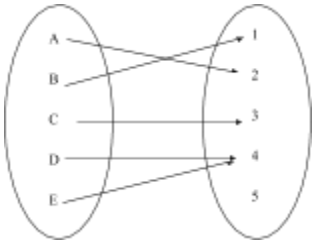
(C)

Input (x)	Output (y)
-1	1
0	1
7	10
3	-1

(D)



(E)



NOTATION:

In most cases, we use f(x) to denote a function of x. Keep in mind that *f* is not a variable; rather it is just the particular notation used traditionally for writing functions.

f(x) is read as "f of x" -- this should not be thought of as *f* multiplied by *x (this is really important!)*
Functions are traditionally represented by single letter names -- g(x), f(x), h(x) all represent functions of x. Sometimes, different letters (i.e. *f* and g) are used to denote different functions, to prevent confusion.
For example:

$f(x) = 3x + 1$
$g(x) = 2x^2 + 7x + 6$

SIDENOTE: f(x) v. *y*

A function is an expression, whereas an equation is 2 expressions set equal to each other.

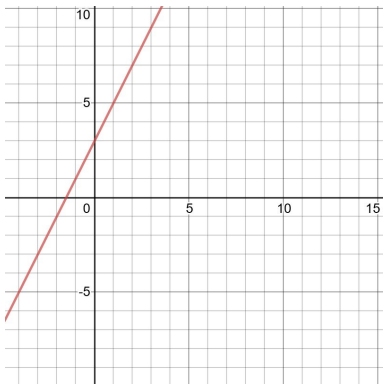
$f(x) = 2x + 1$ defines a function of x: you put in "x", and you get "2x+1".

$y = 2x + 1$ is an equation, in which y is equal to "2x+1". This form is more applicable to graphing functions.

Although we will use both terms interchangeably. Keep in mind that they do not technically mean the same thing.

GRAPHING FUNCTIONS:

We can also plot functions on graphs. On a Cartesian plane (the x-y coordinate plane), we let the x-axis represent the domain and the y- axis the range.
Thus, the y-axis represents the output values -- in other words, $y = f(x)$



Take a look at the function to the right:
$f(x) = 2x + 3$

You can see that every solution (x,y) that satisfies the function above are plotted on the graph. In a sense, the graph represents a set of all solutions to f(x).

We can see that (2,7) is on the graph, since
$7 = 2(2) + 3$
$\Rightarrow 7 = 7$
However, (2,8) does not, since
$8 = 2(2) + 3$
$\Rightarrow 8 \neq 7$

Exercise:
Graph the following functions on a seperate sheet of paper. We suggest that you "plug in" a set amount of x-values (inputs) to get the corresponding output, and connect the points.

1. $f(x) = x^3 + 5x^2 - 7$
2. $g(x) = -3x + 5$
3. $f(x) = -2^x - 15$
4. $y(x) = 7$
5. $g(x) = 3x - 6$
6. $h(x) = -3(x + 5) - 2$
7. $v(x) = -3(x - 3)^2$

<u>DOMAIN:</u>
The domain is the set of all values that the function can take. For instance, let $f(x) = x$. The domain for $f(x)$ is all real numbers ($\mathbb{R}$), since x, the input, can take any value within the set of all real numbers.

This isn't always the case, however. For example, $f(x) = \sqrt{x}$ cannot take any values less than 0, since on a real plane, we cannot take the square root of a negative number. Therefore, the domain of $f(x) = \sqrt{x}$ is $[0, +\infty)$.

*Quick side lesson: Intervals that look like this: [-2, 5] or (-200, 500) are in what we call interval notation. This is basically just an inequality. If x was on the interval [-2, 5], that would mean $x \geq -2$ and $x \leq 5$. The interval just represents all the points from x = -2 to x = 5. Closed brackets **[]** mean a number is included in the interval, while open brackets **()** mean a number is not included in the interval (or is an infinite value: $+\infty$ or $-\infty$).*

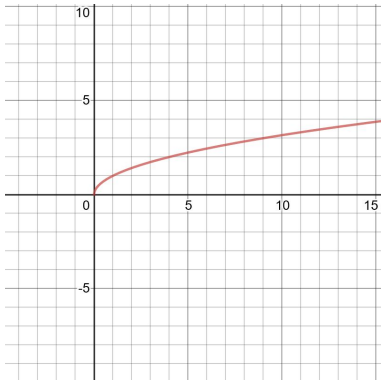
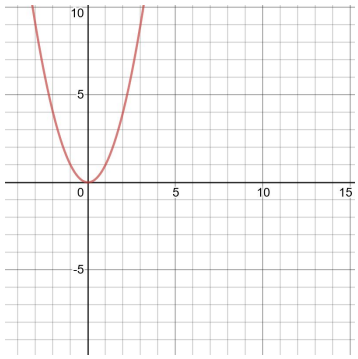
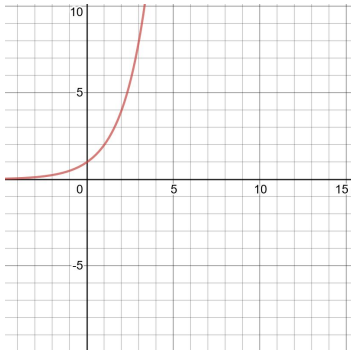
The domain is typically associated with the x-axis; that is, the domain for a function is a set of values on the x-axis.

<u>RANGE:</u>
The range is the set of all outputs of the function. Let $f(x) = x$. The range for f(x) is all real numbers, since the output can be any number within the set of real numbers. $f(x) = x^2$, however, cannot output a negative

number, since the square of any real number is bound to be positive. Thus, the range of $f(x) = x^2$ is $[0, +\infty)$. Likewise, since the square root of any number within the set $[0, +\infty)$ will be positive, the range for $f(x) = x^{1/2}$ is likewise $[0, +\infty)$.

The range is typically associated with the y-axis; that is, the range for a function is a set of values on the y-axis.

For the diagrams below, we denote the domain of function f as Dom (f) and the range as Rng (f)

	$f(x) = x$ Dom $(f) = [0, +\infty)$ Rng $(f) = [0, +\infty)$
	$f(x) = x^2$ Dom $(f) = \mathbb{R}$ Rng $(f) = [0, +\infty)$
	$f(x) = 2^x$ Dom $(f) = \mathbb{R}$ Rng $(f) = (0, +\infty)$

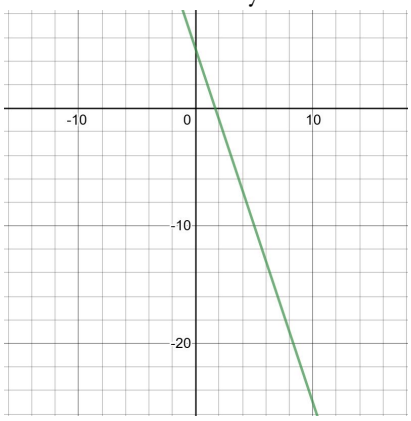
Exercise: Determine the domain and range of the following functions. Graph if necessary.

1. $f(x) = x^3$

2. $g(x) = 3x^2$
3. $y(x) = x + 4$
4. $f(x) = 3(2)^x$
5. $f(x) = (x-1)^{1/2}$
6. $y(x) = 7$
7. $g(x) = 3x - 6$
8. $h(x) = -3(x+5) - 2$
9. $v(x) = -3(x-3)^2$

<u>LINEAR FUNCTIONS:</u>

A linear function is any function that takes the shape of a straight line. For example, this is a linear function:



Here are some examples of linear functions:

$f(x) = x + 5$

$f(x) = 3x$

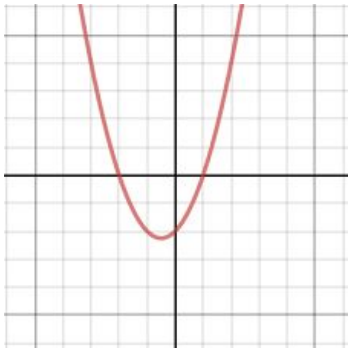
$f(x) = 5(x-3) + 2$

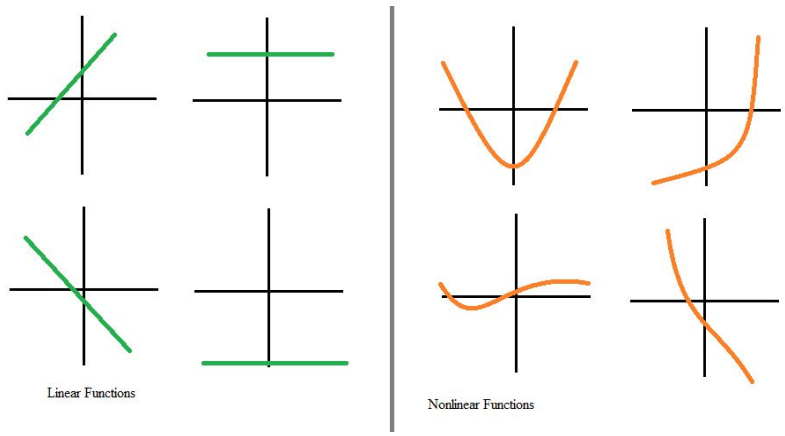
The commonality between all of these is that they all have x raised to the 1st power, x^1 which equals just x.

There is a special case: $f(x) = 3$ is a horizontal line located on all the points with a y-coordinate of 3. Another special case is $x = 5$, which is a vertical line located on all the points with a x-coordinate of 5. (This is technically not a function, though, because it fails the Vertical Line Test.)

<u>NONLINEAR FUNCTIONS:</u>

Nonlinear functions will be increasing or decreasing at an inconsistent rate and will show a curved graph. For example, they could be functions that carry exponents greater than 1 on their first variable in standard form $(ax^2 + bx + c)$.





<u>SLOPE:</u>

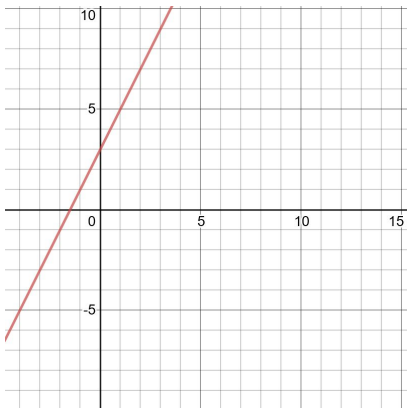
The slope of 2 points on a graph represent the rate of change, or mathematically, the ratio between the change in y and the change in x -- you might have learned this as "rise over run", which encapsulates this concept perfectly. Simply put, when asked to calculate the slope between 2 points of a function, you are asked to find the change in y over the change in x, which can be solved by the following formula:

$$m = \frac{f(b) - f(a)}{b - a}$$

Where m represents the slope, and a and b are elements in the domain (or x-values) that correspond to $f(a)$ *and* $f(b)$*, respectively.*

The slope of a linear function will always remain constant -- that is, the change in y over change in x for any 2 points on the graph will have a constant value.

! NOTE: THE ABOVE FUNCTIONS ARE NOT DRAWN TO SCALE



$f(x) = 2x + 3$

Let's take any 2 points that we know are on the graph.

Let a = 0

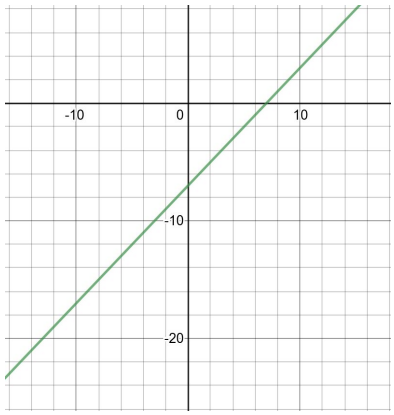
Let b = 1

$f(0) = 2(0) + 3 = 3$

$f(1) = 2(1) + 3 = 5$

$$slope = \frac{f(b) - f(a)}{b - a} = \frac{f(1) - f(0)}{1 - 0} = \frac{5 - 3}{1} = \frac{2}{1} = 2$$

So, the slope of this function is simply 2. Intuitively, if x increases by 1 unit, y will increase by 2 units



$f(x) = 5x-7$

Again, let's take any 2 points that we know are on the graph.

Let a = 0

Let b = 1

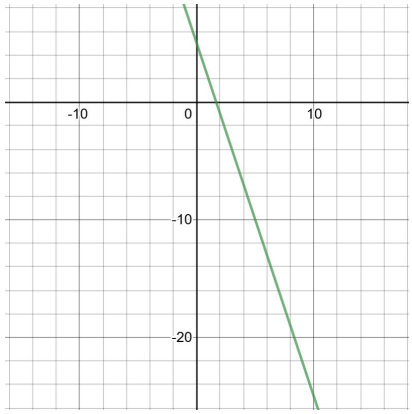
$f(0) = 5(0) -7 = -7$

$f(1) = 5(1) -7 = -2$

$$slope = \frac{f(b) - f(a)}{b - a} = \frac{f(1) - f(0)}{1 - 0} = \frac{-2 - (-7)}{1} = \frac{+5}{1} = 5$$

So, the slope of this function is simply 5.

Furthermore, it doesn't matter what points we plug in for x, as long as both points are within the domain, or defined at x. We will demonstrate this in the next example.



$f(x) = -3x - 5$

This one is just a little bit different, since we have a negative slope. Regardless, we can still find the slope the same way.

Let a = -3

Let b = 5

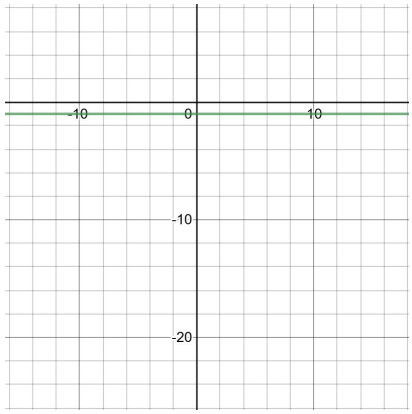
$f(5) = -3(5) - 5 = -20$

$f(-3) = -3(-3) - 5 = 4$

$$slope = \frac{f(b) - f(a)}{b - a} = \frac{f(5) - f(-3)}{5 - (-3)} = \frac{-20 - 4}{5 - (-3)} = \frac{-24}{8} = -3$$

So, the slope of this function is simply -3. Starting to see a pattern?

When written in the form f(x) = mx+b, the slope of the function will simply be *m*. This is called *slope-intercept* form, and we will be

	discussing it shortly.
	$f(x) = -1$ This is a constant function -- for any x, the corresponding y-value will always be -1 *Let a = -7.3* *Let b = 10* $f(10) = -1$ $f(-7.3) = -1$ $slope = \frac{f(b)-f(a)}{b-a}$ $= \frac{f(10)-f(-7.3)}{10-7.3}$ $= \frac{-1-(-1)}{10-7.3}$ $= \frac{0}{10-7.3} = 0$ The slope of any constant function will always be 0. Think about it this way: For every incremental change in x, f(x) doesn't change -- it remains at a fixed value (which in this case is -1).

! When calculating the slope of a function using points [a, f(a)], [b,f(b)], it doesn't matter whether a or b comes first, as long as f(a) and f(b) follow correspondingly. In other words,

$$\frac{f(b)-f(a)}{b-a} = \frac{f(a)-f(b)}{a-b}$$

We can easily show this my manipulating the equation algebraically:

$$\frac{f(b)-f(a)}{b-a} = \frac{f(a)-f(b)}{a-b}$$

$$[f(b) - f(a)]\,[a-b] = [f(a) - f(b)]\,[b-a]$$

$$[f(b) - f(a)]\,[a-b] = -\,[-f(a) + f(b)]\,[b-a]$$

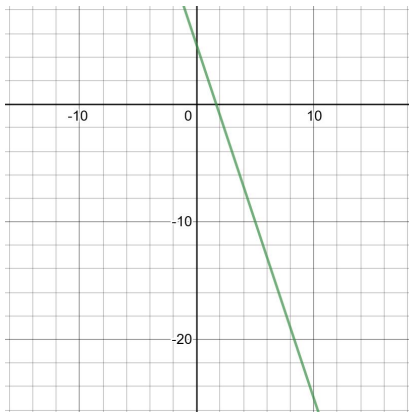
In Steps (2) and (3), we simply factored out the negative sign

$$[f(b) - f(a)]\,[a-b] = +\,[-f(a) + f(b)]\,[-b+a]$$

Which is equivalent to

$$[f(b) - f(a)]\,[a-b] = +\,[f(b) - f(a)]\,[a-b]$$

For instance, let's revisit one of the functions above:



$f(x) = -3x - 5$

Let a = -3

Let b = 5

$f(5) = -3(5) - 5 = -20$

$f(-3) = -3(-3) - 5 = 4$

$$slope = \frac{f(a) - f(b)}{a - b} = \frac{f(-3) - f(5)}{-3 - 5} = \frac{4 - (-20)}{-8} = \frac{24}{-8} = 3$$

Our previous solution is as follows (scroll up if you want to confirm):

$$slope = \frac{f(b) - f(a)}{b - a} = \frac{f(5) - f(-3)}{5 - (-3)} = \frac{-20 - 4}{5 - (-3)} = \frac{-24}{8} = -3$$

To summarize,

$$\frac{f(b) - f(a)}{b - a} = \frac{f(a) - f(b)}{a - b}$$

In other words, it doesn't matter which value we assign to be b *and* a, *as long as the corresponding f(*b*) and f(*a*) match*

There are 2 common methods of writing a linear function that you should know. We will briefly review them here.

Slope-Intercept Form:

$y = mx + b$ *; where m is the slope and b is the y intercept (the point where the function crosses the y-axis)*

**The x-intercept is the point where the graph crosses the x axis*

This form is usually used when one has directly been given 2 coordinates (x_1, y_1) (x_2, Y_2)

As you might have noticed by now, all functions shown so far in the examples have been written in slope-intercept form

Point-Slope Form

$(Y_2 - y_1) = m(x_2 - x_1)$ *; where m is the slope and* y_1 *and* x_1 *represent a known coordinate* (x_1, y_1) *on the function.*

This form is usually used when one has directly been given 1 coordinate and the slope.

It is easy to go from slope-intercept form to point-slope form, although we will leave this as an exercise for the reader to prove for themselves.